

# The sine and cosine functions and vertical dilations



1 Consider the expression  $\cos \theta$ .

a Complete the table of values for different values of  $\theta$ .

|               |   |                 |                 |                  |       |                  |                  |                  |        |
|---------------|---|-----------------|-----------------|------------------|-------|------------------|------------------|------------------|--------|
| $\theta$      | 0 | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ | $\frac{2\pi}{3}$ | $\pi$ | $\frac{4\pi}{3}$ | $\frac{3\pi}{2}$ | $\frac{5\pi}{3}$ | $2\pi$ |
| $\cos \theta$ |   |                 |                 |                  |       |                  |                  |                  |        |

b Sketch a graph of the function  $y = \cos \theta$ .

c State the maximum value of  $\cos \theta$ .

d State the minimum value of  $\cos \theta$ .

e State the range of values of  $4 \cos \theta$ .

2 Consider the functions  $y = \sin x$  and  $y = \cos x$ , where  $x$  is in radians.

a State the domain of both functions.

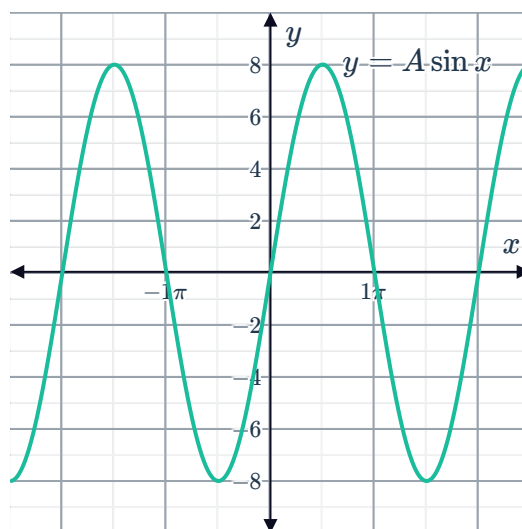
b State the range of both functions.

3 Consider the function  $y = -2 \cos x$ , where  $x$  is in radians.

a State the domain of the function.

b State the range of the function.

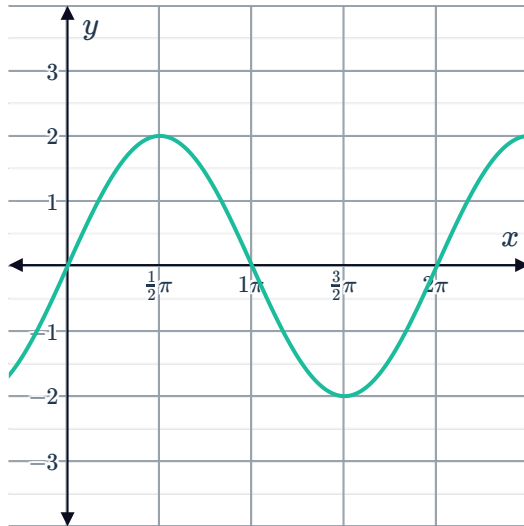
4 Consider the given graph of a function of the form  $f(x) = A \sin x$ :  
State the amplitude of the function.



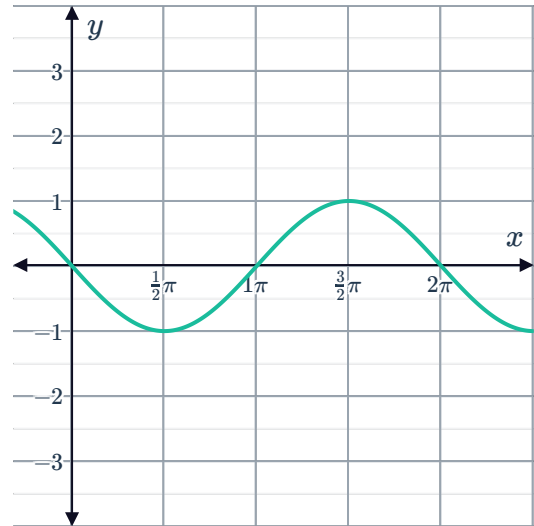
- 5 State the equation of each of the functions sketched below given that they are of the form  $y = a \sin x$  or  $y = a \cos x$ :



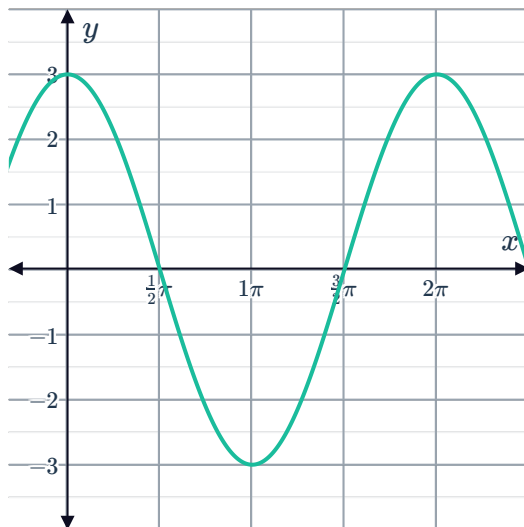
a



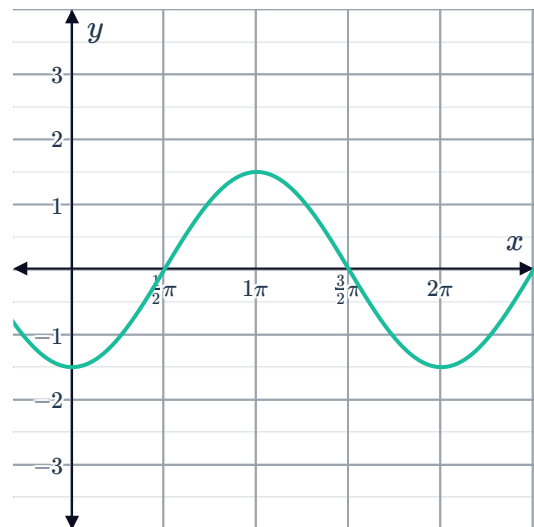
b



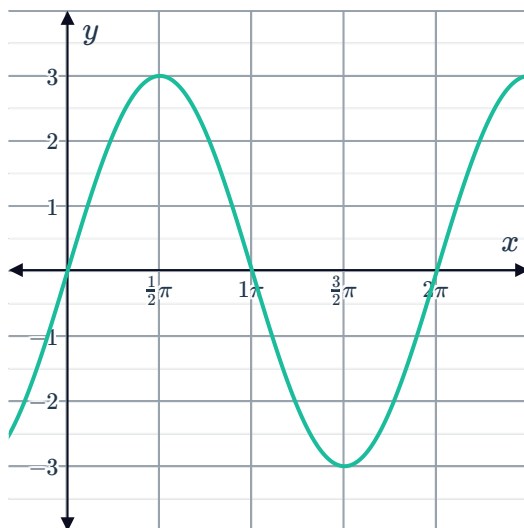
c



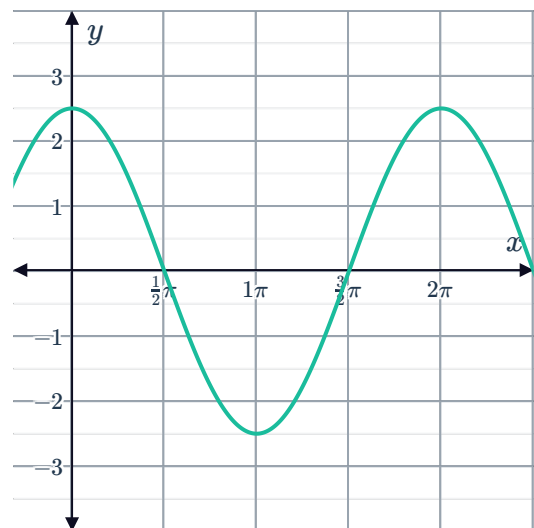
d



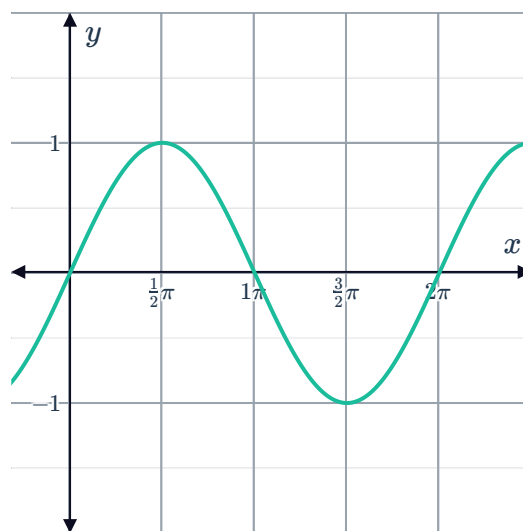
e



f



- 6 Consider the graph of the function  $y = \sin x$  for  $0 \leq x < 2\pi$ :  
At which value of  $x$  in the given domain would  $y = -\sin x$  have a maximum value?

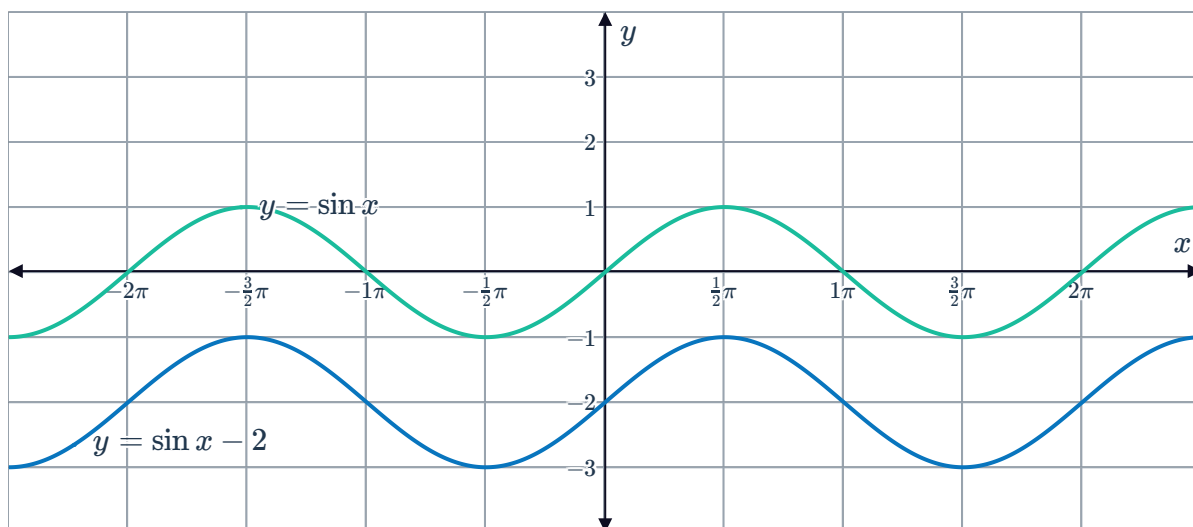


- 7 Consider the function  $f(x) = -8 \sin x$ , where  $0 \leq x \leq \pi$ .
- State the amplitude of the function.
  - Find the value of  $f(\pi)$ .
  - Find the minimum value of the function.
- 8 The function  $y = k \sin x$  has a maximum value of 5. Find the value of  $k$ , where  $k > 0$ .
- 9 A sine function has the form  $y = c \sin x$ , a range of  $[-2, 2]$  and a minimum at  $\frac{\pi}{2}$ . Find an expression for  $y$ .
- 10 For each of following functions:
- State the amplitude.
  - Sketch a graph of the function.
- $y = 3 \sin x$
  - $y = 4 \cos x$
  - $y = \frac{4}{3} \cos x$
  - $y = -5 \cos x$
  - $y = \frac{5}{4} \sin x$
  - $y = -4 \sin x$
- 11 Consider the function  $y = -3 \cos x$ .
- State the maximum value of the function.
  - State the minimum value of the function.
  - State the amplitude of the function.
  - Describe the two transformations required to obtain the graph of  $y = -3 \cos x$  from the graph of  $y = \cos x$ .



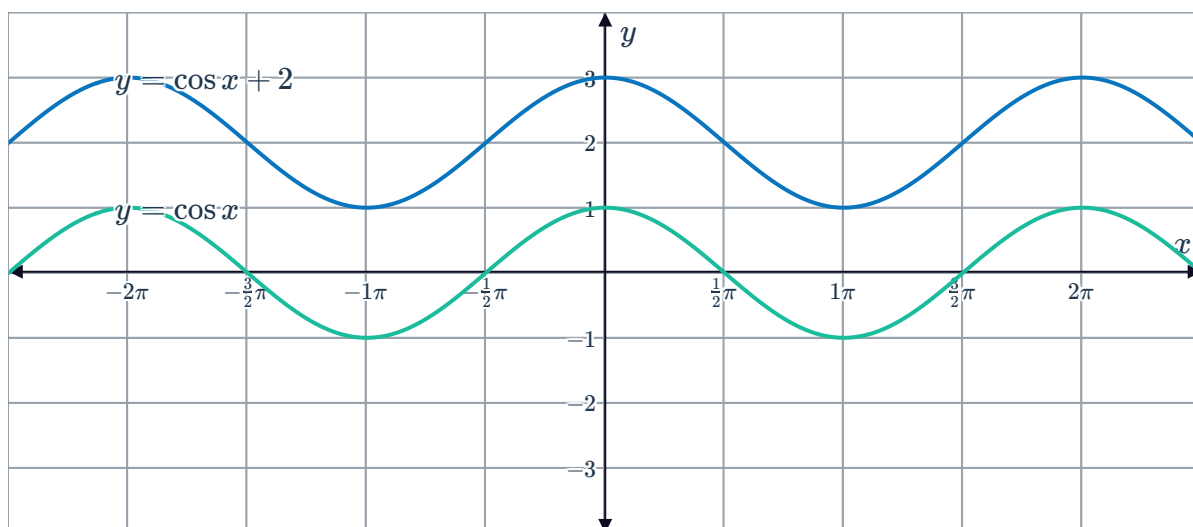
## Vertical translations of sine and cosine functions

12 Consider the two graphs  $y = \sin x$  and  $y = \sin x - 2$  below:



Describe the transformation required to obtain the graph of  $y = \sin x - 2$  from  $y = \sin x$ .

13 Consider the two graphs  $y = \cos x$  and  $y = \cos x + 2$  below:



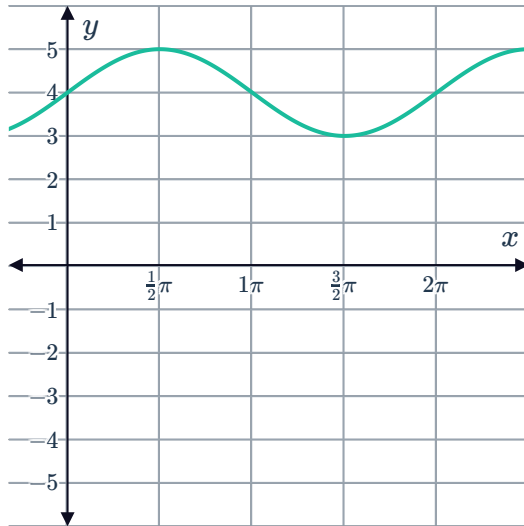
Describe the transformation required to obtain the graph of  $y = \cos x + 2$  from  $y = \cos x$ .

14 Describe the transformation required to obtain the graph of  $y = \sin x + 4$  from  $y = \sin x$ .

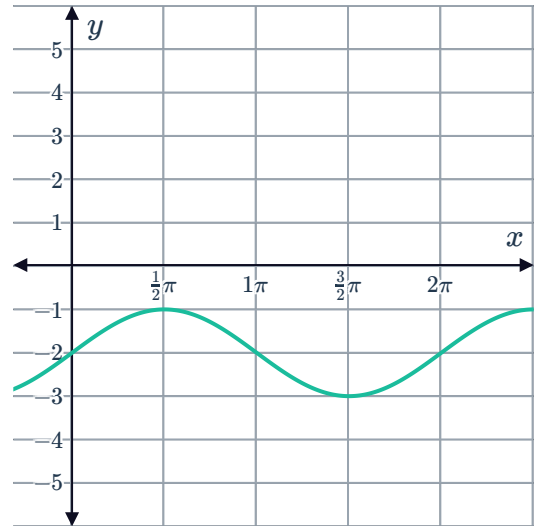
- 15 State the equation of each of the functions plotted below given that they are of the form  $y = \sin x + k$ :



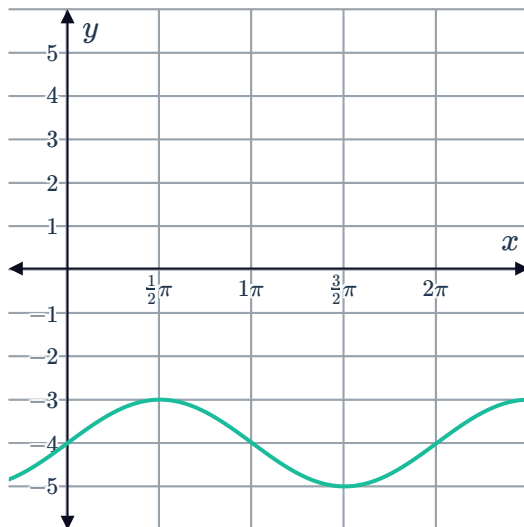
a



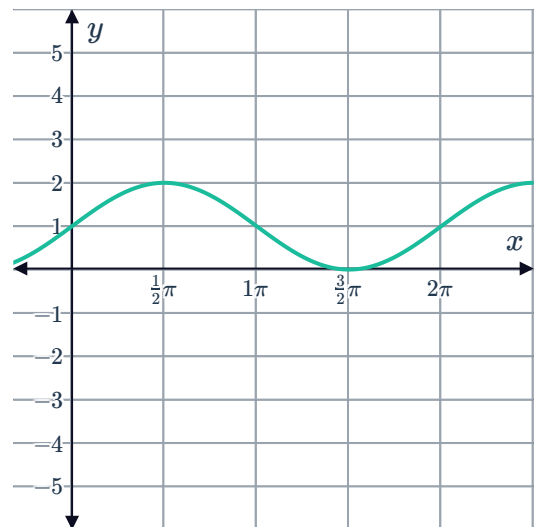
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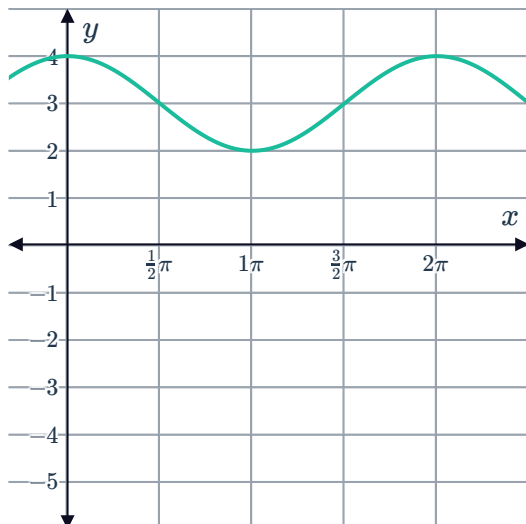


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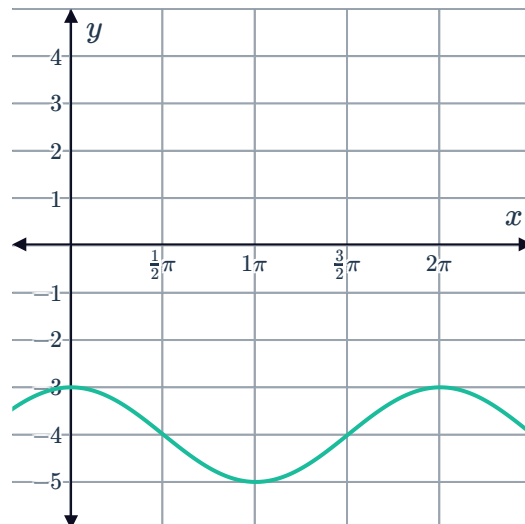


- 16 State the equation of each of the functions graphed below given that they are of the form  $y = \cos x + k$ :

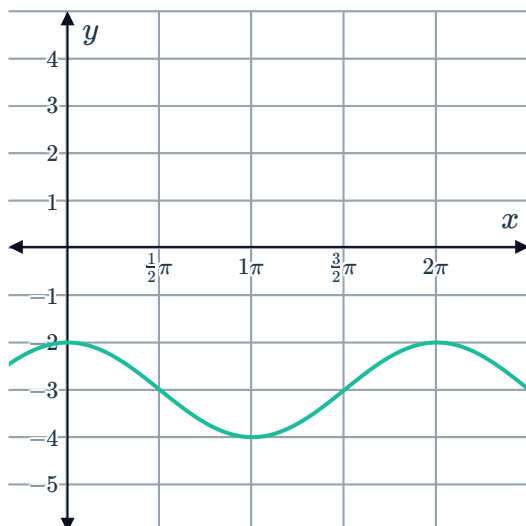
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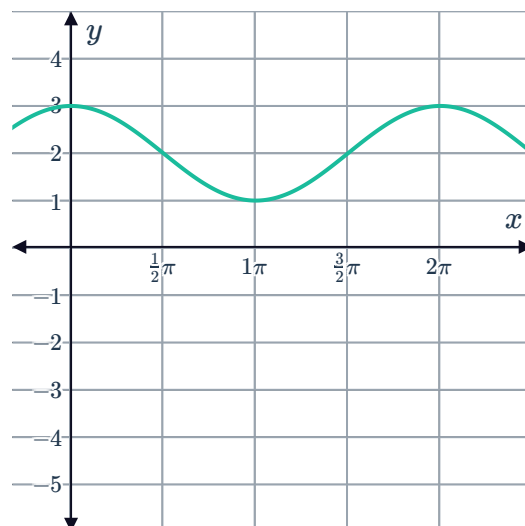
b



c



d



- 17 Describe the two transformations required to turn the graph of  $y = \cos x$  into the graph of  $y = -\cos x + 3$ .

- 18 State the equation of the line that the function  $y = \sin x + 2$  oscillates about.

- 19 Consider the function  $y = \cos x + 4$ .

- Describe the transformation required to obtain the graph of  $y = \cos x + 4$  from  $y = \cos x$ .
- State the period of the function, giving your answer in degrees.
- Find the maximum value of the function.
- Find the minimum value of the function.

**20** For each of the functions below:



- i State the period of the function, giving your answer in radians.
- ii State the amplitude of the function.
- iii Find the maximum value of the function.
- iv Find the minimum value of the function.
- v Sketch a graph of the function for  $-2\pi \leq x \leq 2\pi$ .

**a**  $y = \sin x + 1$       **b**  $y = \cos x - 3$       **c**  $y = \sin x - 2$       **d**  $y = 3 \sin x + 2$   
**e**  $y = 3 \cos x - 3$       **f**  $y = 2 \sin x - 3$

**21** For each of the functions below:

- i Find the value of  $y$  when  $x = \frac{\pi}{2}$ .
- ii State the amplitude of the function.
- iii Find the maximum value of the function.
- iv Find the minimum value of the function.
- v Describe the transformations required to obtain the graph of the function from  $y = \cos x$ .
- vi Sketch a graph of the function for  $0 \leq x \leq 2\pi$ .

**a**  $y = 2 \cos x + 3$       **b**  $y = 2 \cos x - 3$       **c**  $y = -4 \cos x + 2$       **d**  $y = \frac{1}{3} \sin x + 2$

**22** The function  $y = \cos x + 5$  is translated 4 units up.

- a** State the equation of the new function after the translation.
- b** Find the maximum value of the new function.

**23** For each of the functions below:

- i Find the maximum value of the function.
- ii Find the minimum value of the function.

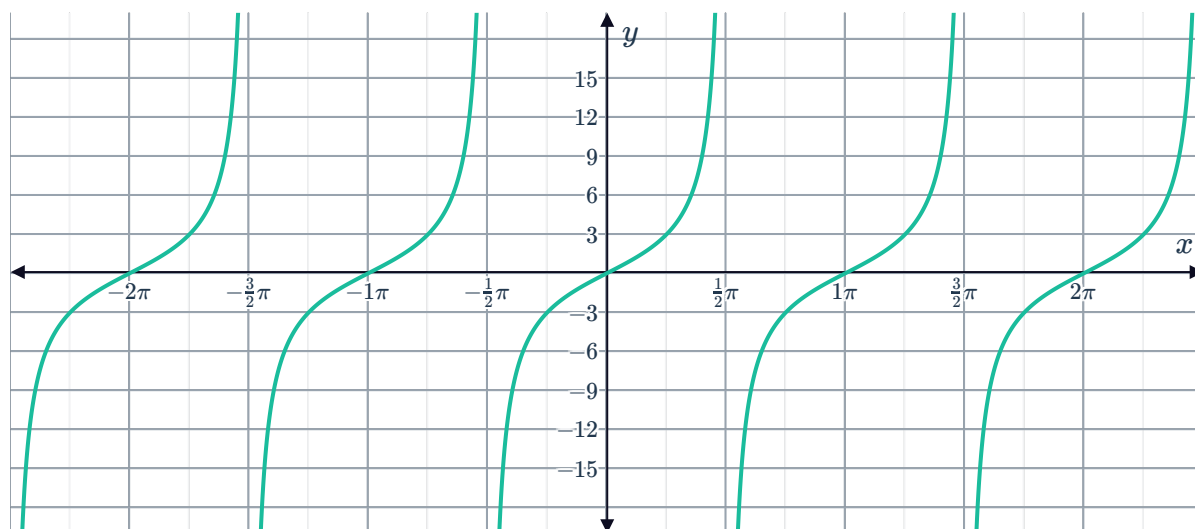
**a**  $y = 2 - 3 \sin x$       **b**  $y = 4 - 3 \sin x$       **c**  $y = (\sin x)^2 + 1$       **d**  $y = \frac{4}{3 \cos x + 5}$

**24** A sine function,  $y$ , has the form  $y = c \sin x + d$  and a range of  $[0, 4]$ . Find an expression for  $y$ , where  $c > 0$ .

**25** A cosine function,  $y$ , has the form  $y = c \cos x - d$  and a range of  $[-10, 6]$ . Find an expression for  $y$ , where  $c > 0$ .

## The tangent function and vertical dilations

26 Consider the given graph of  $f(x) = 3 \tan x$ :

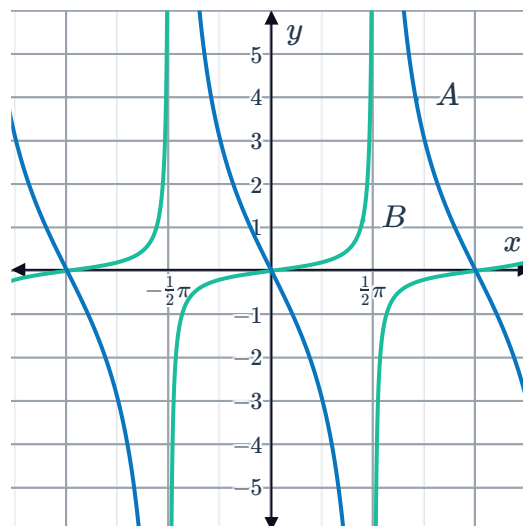


- State the domain of  $f(x)$ .
- State the range of  $f(x)$ .
- The graph of  $f(x)$  has its domain restricted to  $[0, \pi)$ , state the range of the restricted graph.

27 Consider the functions  $f(x) = \frac{1}{5} \tan x$  and  $g(x) = -3 \tan x$  drawn on the given set of axes:

Which function has been labelled:

- A
- B



28 For each of the functions below:

- State the vertical dilation factor required to obtain the given function from  $y = \tan x$ .
- State whether the graph would appear stretched or compressed in relation to  $y = \tan x$ .

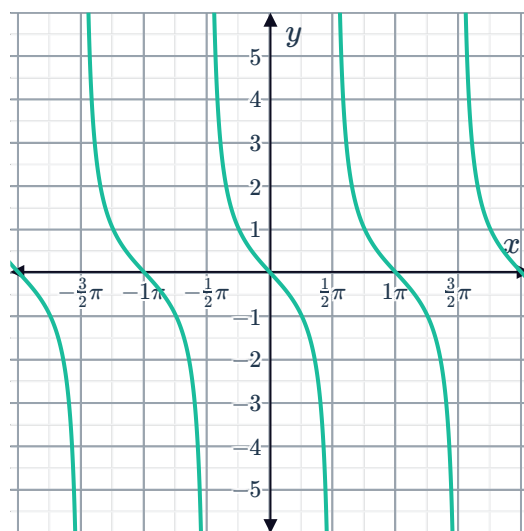
- $y = \frac{1}{4} \tan x$
- $y = 6 \tan x$
- $y = \frac{5}{3} \tan x$
- $y = 0.8 \tan x$



29 Consider the given graph of a function of the form  $y = a \tan x$ :

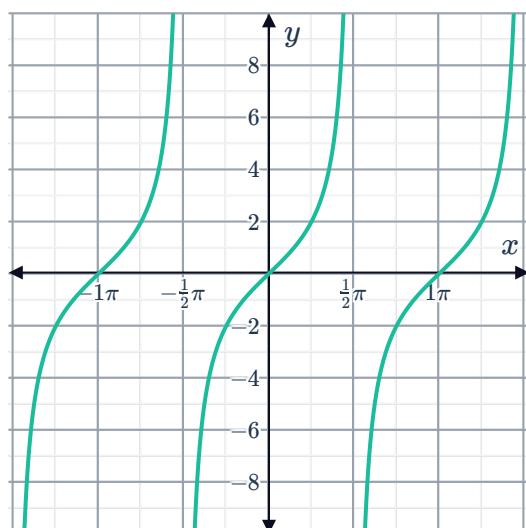


- Describe the transformation required to obtain the given function from  $y = \tan x$ .
- State the equation of the graphed function.

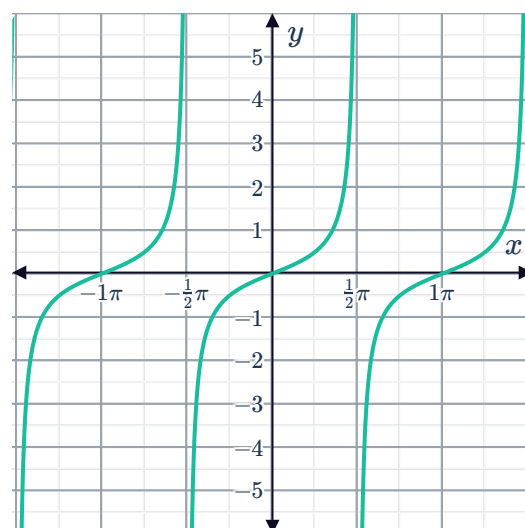


30 State the equation of each of the functions graphed below given that they are of the form  $y = a \tan x$ :

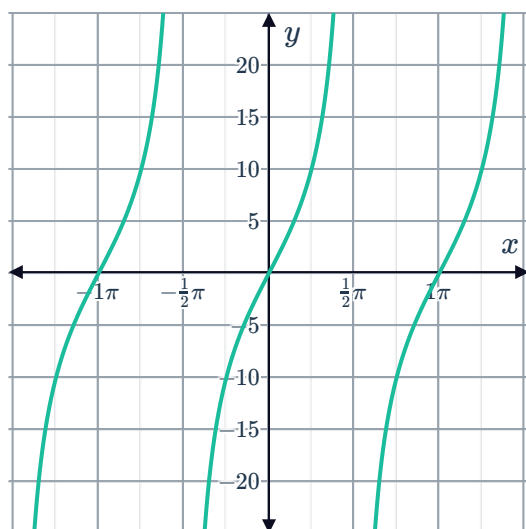
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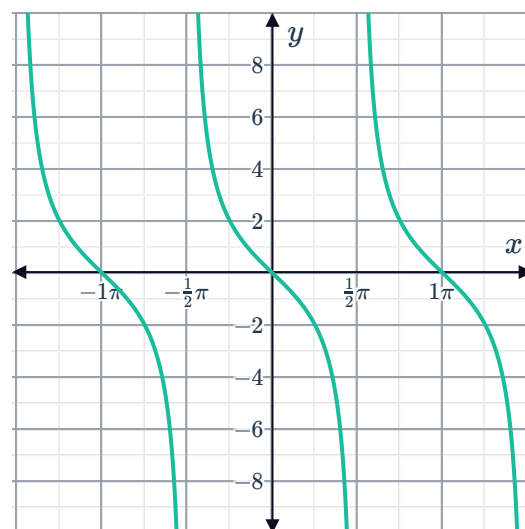
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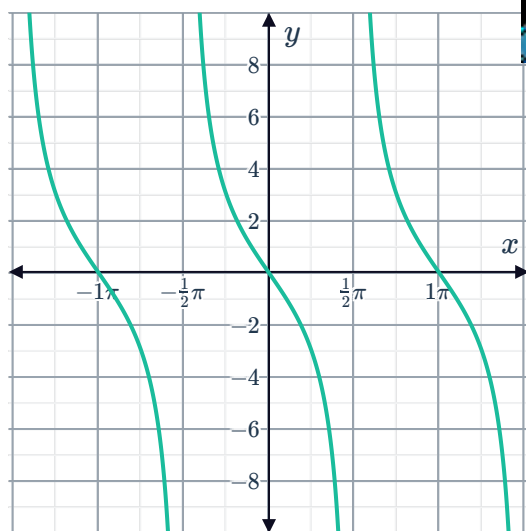
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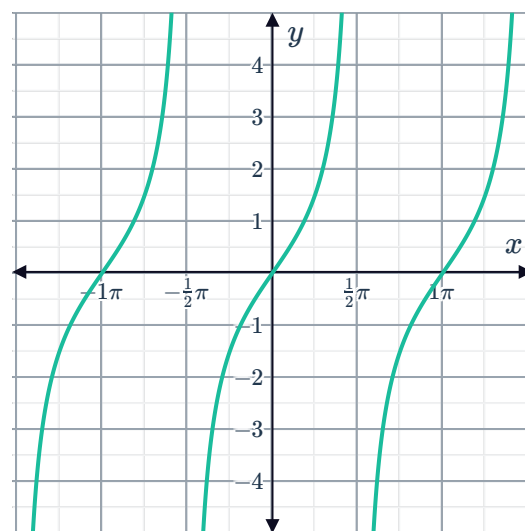
d



e



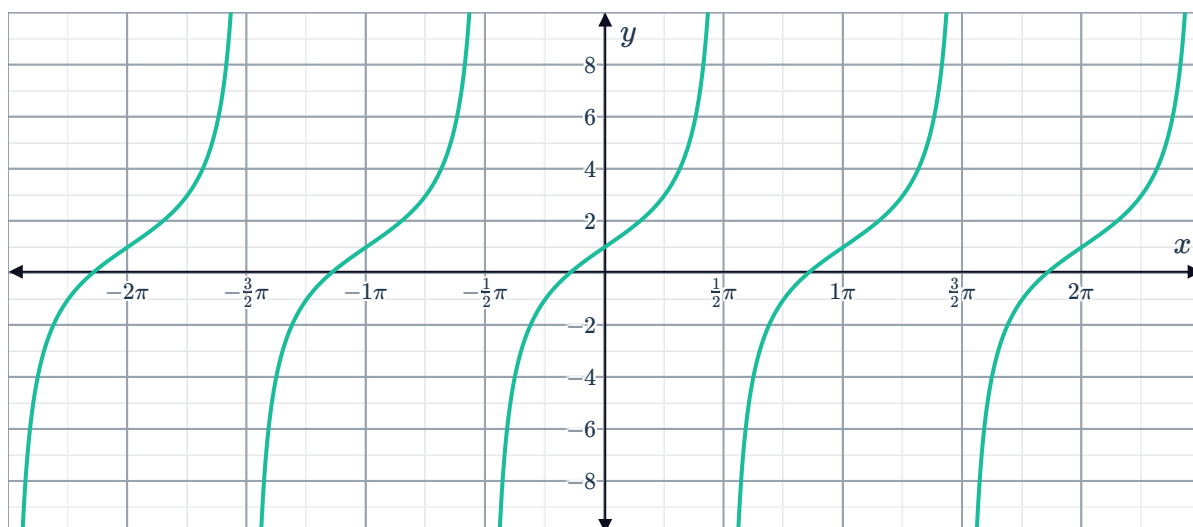
f



- 31 On the same set of axes, sketch the graph of  $y = \tan x$  and  $y = \frac{1}{2} \tan x$ .

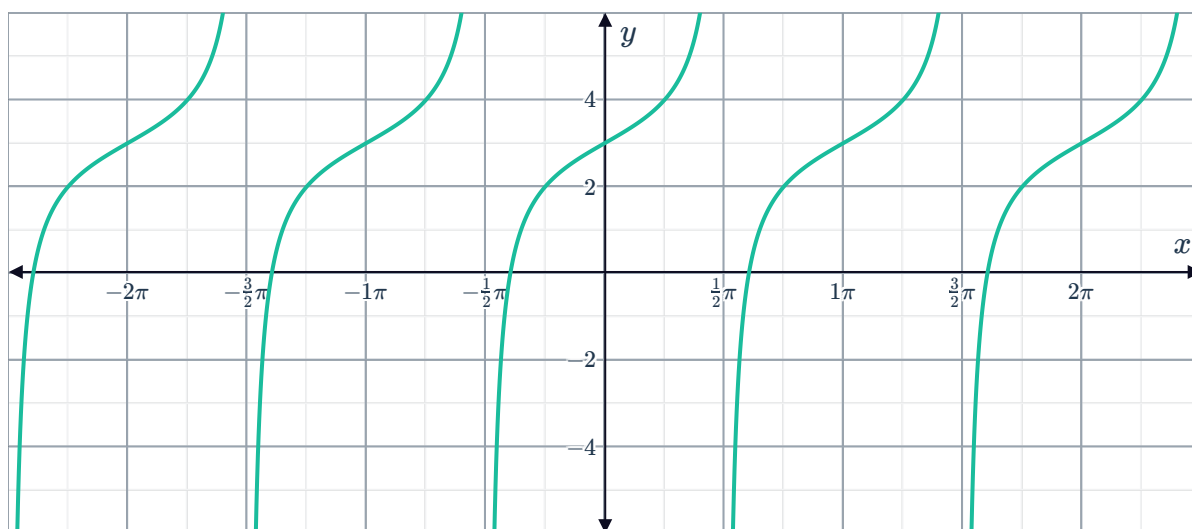
## Vertical translations of the tangent function

- 32 Consider the given graph of  $f(x) = 2 \tan x + 1$ :



- a State the domain of  $f(x)$ . b State the range of  $f(x)$ .

**33** Consider the given graph of  $f(x) = \tan x$



- a** State the domain of  $f(x)$ .
- b** State the range of  $f(x)$ .
- c** The graph of  $f(x)$  has its domain restricted to  $\left[0, \frac{\pi}{2}\right)$ , state the range of the restricted graph.

**34** For each of the functions below:

- i** Find the  $y$ -intercept.
- ii** Find the value of  $y$  when  $x = \frac{\pi}{4}$ .
- iii** Find the period of the function.
- iv** Find the distance between the asymptotes of the function.
- v** State the equation of the first asymptote of the function for  $x \geq 0$ .
- vi** State the equation of the first asymptote of the function for  $x \leq 0$ .
- vii** Sketch a graph of the function.

**a**  $y = \tan x - 2$

**b**  $y = 5 \tan x + 3$

**35** Consider the given graph of a tangent function:



- a** State the transformation required to obtain the given function from  $y = \tan x$
- b** State the equation of the graphed function.

